

James Lavis

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Summary

Data Scientist with 3+ years of experience developing and implementing machine learning models, data mining techniques, graph analytics, and statistical analysis to generate prescriptive and predictive insights in both scientific and corporate environments. Self-motivated and passionate about leveraging data and advanced analytics to drive business value and informed decision-making.

Experience

Data Scientist at The Australian Taxation Office

November 2022 – Present

Research and implement machine learning, data mining, and graph analytics techniques on big data to enhance intelligence insights for case officers responsible for maintaining the integrity of the Australian taxation system.

Key responsibilities

- Collaborate with stakeholders to understand complex business problems
- Extract, clean, process, and analyze data from a data warehouse using SQL.
- Develop machine learning models and data science solutions for both tabular and graph-structured data using Python (native), Polars, PyTorch, and Scikit-learn.
- Deploy solutions into production environments using GitLab CI/CD, Apache Airflow, and later Kubeflow.

Achievements

- Improved the performance of a key recommender system by 20%, creating a more efficient model that reduced cost and runtime
- Pioneered the development of a novel evaluation metric for an entity resolution algorithm
- Presented methods, findings, and strategic recommendations to key executives at the Australian Taxation Office (ATO).

Student Researcher at The Commonwealth Scientific and Industrial Research Organisation

July 2022 – Feb 2024

Research Component of Master's Degree and Independent Collaboration: Explored the application of deep learning techniques to enhance a key mathematical model for conservation planning.

Key responsibilities

- Conducted an in-depth literature review on Graph Neural Networks (GNNs) and Markov Decision Processes (MDPs)
- Simulated various conservation scenarios to generate MDPs, used as training data for deep learning models.
- Implemented, trained, and evaluated multiple GNN architectures, utilizing techniques such as hyperparameter tuning and cross-validation.

Achievements

- Contributed a key research component for the Master's of Data Science degree.
- Continued collaboration with researchers post-degree, with results informing future research directions at CSIRO.

Research Assistant at The Superconducting Quantum Devices Laboratory (University of Queensland)

July 2020 – Dec 2021

Was responsible for a number of tasks enabling academics to conduct research into building superconducting quantum devices, with eventual application in quantum computing.

Key responsibilities

- Design and simulate components for use in dilution refrigerators operating at temperatures as low as 30 mK.
- Quickly adapt to the laboratory's software stack, enabling efficient collection and analysis of experimental data from microwave electronics and superconducting devices.
- Collect and analyse scientific data from simulations and experiments
- Assist with the setup and operation of delicate electronics and equipment

Achievements

- Resulted in a Bachelor of Science (Honours Class 1) in Physics from the University of Queensland

Education

Master of Data Science from The University of Queensland

Graduated 2023

Bachelor of Science (Honours Class 1) with a major in Physics from The University of Queensland

Graduated 2021

Key skills	Python, R, C++, Data Science, Data Mining, Supervised Learning, Unsupervised Learning, Graph Analytics and Network Science, Recommendation Systems, Data Analytics, Statistical Analysis, A/B Testing, Risk Neutral Pricing, Calculus and Linear Algebra, Statistics, PowerBI, SQL, Spark, MLOps, Airflow, Neo4j, Kubeflow, Linux, Docker, Git, OpenAI LLM's, Prompt Engineering, API's, Microsoft PowerPoint, Microsoft Office, Client Engagement, Business and Process Analysis
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Languages	English - Native speaker German - CEFR-level A1 and improving
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References	Available upon request
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